AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning

R26-SE-018

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

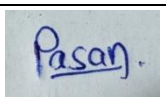
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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Orchid cultivation requires careful control of environmental conditions such as temperature, humidity, light, water supply, and nutrient availability. Among orchid species, Vanda orchids are widely cultivated due to their high commercial value and ornamental characteristics [7], [8]. However, Vanda orchids have exposed roots and require precise watering and nutrient management. Traditional orchid care methods rely mainly on manual observation and fixed watering schedules, which may not reflect the real-time needs of the plant. As a result, plants may receive too much or too little water and fertilizer, negatively affecting plant growth and flowering.

Recent research has introduced IoT-based monitoring systems that use environmental sensors to measure conditions around plants. These systems can collect data such as temperature, humidity, and moisture to support automated irrigation decisions [1], [5]. However, many existing systems still rely on threshold-based irrigation control, where watering is activated only when moisture levels drop below a predefined value [2], [4]. Such systems respond only to current environmental conditions and do not predict future plant requirements. In addition, many existing solutions mainly focus on irrigation and do not integrate automated fertilization mechanisms within the same system.

To address these limitations, this research proposes a Smart Watering and Automated Fertilization System for Vanda Orchids using Multi-Modal Machine Learning. The system collects environmental data using sensors such as temperature, humidity, light intensity, and moisture connected to an IoT device. The collected data is stored and analyzed to understand environmental patterns and plant behavior. ML techniques can then analyze historical environmental data to support more intelligent irrigation decisions [9], [10]. Based on these predictions, the system automatically controls a water pump and a fertilizer pump to deliver the required irrigation and nutrients.

The proposed system aims to improve plant health, reduce water and fertilizer waste, and support more efficient orchid cultivation. Smart monitoring and automated irrigation technologies have been shown to improve agricultural efficiency and reduce resource consumption in precision farming environments [6], [8]. Therefore, the proposed approach can benefit both small-scale orchid growers and commercial orchid farms by enabling smarter plant care through data-driven decision making.

Keywords: Smart irrigation, IoT agriculture, Vanda orchids, automated fertilization, ML.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all the individuals who supported and guided me throughout the completion of this research proposal.

First and foremost, I would like to thank my supervisor for providing valuable guidance, constructive feedback, and continuous encouragement during the development of this research. Their insights and suggestions greatly contributed to improving the quality of this work.

I would also like to extend my appreciation to the academic staff of the Faculty of Computing at SLIIT for providing the necessary knowledge and academic support that helped shape this research.

My sincere thanks go to my project group members for their cooperation, discussions, and teamwork throughout the project development process. Their support made it possible to successfully complete this proposal.

Finally, I would like to express my heartfelt gratitude to my family and friends for their constant encouragement, understanding, and support throughout my academic journey.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
ESP32	Low-power microcontroller with Wi-Fi and Bluetooth
API	Application Programming Interface
DHT	Digital Humidity and Temperature Sensor
DB	Database
UI	User Interface
FR	Functional Requirement
NFR	Non-Functional Requirement
LDR	Light Dependent Resistor
RH	Relative Humidity

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1. INTRODUCTION

1.1.Background and Context

Orchids are one of the most popular ornamental plants in the world and are widely cultivated for their attractive flowers and high market value. Among the many orchid varieties, Vanda orchids are especially valuable because of their bright colors and long-lasting blooms, making them popular in both home gardens and commercial orchid farms [7], [8].

However, maintaining healthy orchid plants requires careful control of environmental conditions. Factors such as temperature, humidity, light intensity, watering frequency, and nutrient supply directly affect plant growth and flowering. If these conditions are not properly maintained, plants may experience slow growth, weak root systems, or poor flowering performance.

Vanda orchids are epiphytic plants, meaning they grow without soil and have exposed aerial roots. Because of this structure, they are highly sensitive to incorrect watering and fertilization. Excess water can cause root rot, while insufficient watering can lead to dehydration and plant stress. Similarly, improper fertilizer application can lead to nutrient imbalance and damage plant health.

In many orchid farms and home gardens, watering and fertilization are still performed manually by growers. Farmers often follow fixed schedules such as watering plants once or twice per day. However, plant water requirements change depending on environmental conditions such as temperature, humidity, and sunlight. For example, plants may require more water during hot and dry conditions and less water during cooler conditions. Because manual watering schedules do not consider these environmental variations, plants may receive either excessive or insufficient water.

Recent developments in smart agriculture and IoT (IoT) technologies have introduced automated plant monitoring systems. These systems use sensors to measure environmental conditions and allow farmers to monitor plant environments in real time [5], [6]. Some automated irrigation systems activate watering when soil moisture drops below a predefined threshold level [1], [2]. While such systems improve irrigation efficiency, many still rely on rule-based control mechanisms rather than intelligent prediction models.

Another limitation is that many existing smart irrigation systems focus only on watering automation and do not integrate automated nutrient management within the same system. Fertilizer is often applied manually based on the grower’s experience rather than data-driven decision making. Smart irrigation research in precision agriculture suggests that combining environmental sensing with intelligent data analysis can significantly improve water and nutrient management efficiency [8], [9].

To address these challenges, this research proposes a Smart Watering and Automated Fertilization System for Vanda Orchids using Multi-Modal ML. The system will use IoT sensors to collect environmental data such as temperature, humidity, light intensity, and moisture levels. The collected data will be analyzed using ML techniques to predict future watering needs. Previous studies have demonstrated that ML models can support more accurate irrigation decision making compared with simple threshold-based systems [9], [10]. Based on these predictions, the system will automatically control watering and fertilization through connected pumps.

This approach will help improve plant health, reduce water and fertilizer waste, and support more efficient orchid cultivation.

Sustainable Development Goal Alignment

This research aligns with Sustainable Development Goal 12: Responsible Consumption and Production. Efficient management of water and fertilizer resources can help reduce resource waste and promote sustainable agricultural practices.

The system also contributes to Sustainable Development Goal 9: Industry, Innovation and Infrastructure by introducing innovative smart agriculture technology that supports data-driven farming practices.

Research Cluster Alignment

This research belongs to the AI and Smart IoT Systems research cluster, as it integrates ML, IoT devices, and environmental monitoring technologies to support intelligent decision making in agriculture.

1.2.Literature Review

Smart agriculture technologies have recently been introduced to improve plant monitoring and cultivation management. With the development of IoT devices and ML techniques, researchers have proposed several automated systems that can monitor environmental conditions and assist farmers in managing plant growth. These systems typically use environmental sensors to measure parameters such as temperature, humidity, soil moisture, and light intensity, allowing farmers to monitor plant environments in real time and support more efficient crop management [5], [6].

Recent research has also explored intelligent systems for orchid cultivation and monitoring. Advanced smart agriculture solutions combine environmental sensing, image analysis, and automated control systems to improve plant growth monitoring and cultivation management. For example, smart greenhouse monitoring systems can analyze plant conditions and environmental data to support precision agriculture and improve plant health [3], [7].

Several studies have specifically focused on automated irrigation systems for plants and crops. IoT-based irrigation systems commonly use soil moisture sensors and

microcontrollers to monitor plant conditions and activate watering when moisture levels fall below predefined thresholds. These systems can improve irrigation efficiency and reduce manual work for farmers. However, most of these systems rely on rule-based control methods rather than intelligent prediction models [1], [9].

One study proposed an IoT-based automatic watering system for Papuan Black Orchids, where soil moisture sensors were used to detect moisture levels around the plant roots. When the moisture value dropped below a certain threshold, a water pump was automatically activated. Although this approach improved irrigation automation, the system only reacted to current environmental conditions and did not predict future plant water requirements [1].

Another study investigated automated water distribution systems for orchid cultivation using environmental monitoring and threshold-based irrigation control. Water was supplied when the soil moisture level dropped below a predefined value. While the study showed improvements in irrigation efficiency and plant growth, irrigation decisions were still based on fixed threshold rules and did not use predictive models for irrigation planning [2].

The OrchiZen Hybrid Integrated Smart Farming System introduced an integrated smart farming platform for orchid plantations that used IoT sensors and computer vision techniques to monitor plant conditions and provide fertilizer recommendations. However, fertilizer application was not automated and still required manual intervention from the user [3].

Another study developed a monitoring system that predicts room humidity for orchid cultivation environments using temperature and humidity sensors together with a regression model. When the predicted humidity dropped below a certain level, a mist maker was activated to increase humidity. Although the system helped maintain environmental conditions, it focused mainly on humidity control rather than predicting plant watering requirements [4].

Overall, these studies demonstrate that IoT-based monitoring systems can improve plant management and irrigation efficiency. However, many existing systems still rely on threshold-based control methods, limited environmental data, and manual nutrient management. Therefore, there is a need for intelligent systems that combine environmental sensing, predictive models, and automated fertilization to support more efficient orchid cultivation.

1.3.Research Gap

Although several IoT-based monitoring systems have been developed for plant irrigation and environmental monitoring, many limitations still exist in current research. Most existing systems use reactive threshold-based irrigation methods, where watering is

activated only when moisture or humidity levels fall below predefined thresholds. While this approach automates irrigation, it only reacts to current environmental conditions and does not consider historical data or environmental trends [1], [2].

Another limitation is the lack of automated nutrient management. Some smart farming systems provide fertilizer recommendations based on plant characteristics or growth stages, but fertilizer application is still performed manually rather than through automated mechanisms integrated with irrigation systems [3].

In addition, several monitoring systems rely on limited environmental parameters, such as soil moisture or humidity alone. However, plant water requirements are influenced by multiple environmental factors including temperature, humidity, light intensity, and plant growth conditions. Systems that depend on a single parameter may therefore fail to accurately represent the real needs of the plant [4].

Furthermore, most existing research focuses on general crop irrigation systems or other orchid species, while limited studies address intelligent irrigation systems specifically designed for Vanda orchids, which have exposed root systems and unique water requirements. Some studies have explored advanced sensing approaches to monitor plant conditions, but they do not provide automated irrigation or fertilization control mechanisms [12].

To address these limitations, this research proposes a Smart Watering and Automated Fertilization System for Vanda Orchids using Multi-Modal ML. The proposed system integrates environmental monitoring, predictive analytics, and automated plant care mechanisms to improve plant health while reducing water and fertilizer waste.

The main research contributions of this study are:

- Development of a multi-modal monitoring system that collects environmental data such as temperature, humidity, light intensity, and root moisture.
- Use of historical environmental data and ML models to predict future watering requirements instead of relying on simple threshold-based control.
- Integration of automated irrigation and automated fertilization mechanisms using water and fertilizer pumps.
- Implementation of a predictive plant care system that can automatically respond to environmental changes.
- Focus on Vanda orchids, which are less studied in intelligent irrigation research compared to other orchid species.

By integrating environmental monitoring, predictive analytics, and automated plant care mechanisms, the proposed system aims to improve plant health while reducing water and fertilizer waste.

Table 1 : Research Gap Comparison

Research Paper	Approach	Technologies Used	Limitations
Internet of Things on Automatic Watering Systems for Papuan Black Orchids	Developed an IoT-based automatic watering system that activates irrigation when soil moisture falls below a threshold.	Soil moisture sensors, microcontroller, IoT monitoring platform	Uses only threshold-based watering. Does not predict future watering needs. Does not include nutrient management.
Analyzing the Influence of Automated Water Distribution Systems on Precision Irrigation for Orchids	Proposed an automated irrigation system that monitors environmental conditions and activates watering when soil moisture drops below a set level.	Soil moisture sensor, temperature sensor, humidity sensor, IoT monitoring system	Irrigation decisions are reactive. No predictive model is used. Fertilization management is not included.
OrchiZen Hybrid Integrated Smart Farming System for Orchid Plantations	Developed a smart orchid farming system using IoT sensors and computer vision to monitor plants and recommend fertilizers.	Environmental sensors, ESP32, ML models, YOLO object detection	Provides fertilizer recommendations but does not automate fertilizer delivery. Watering decisions are not predictive.
Implementation of Monitoring System and Prediction of Room Humidity for Orchid	Developed a monitoring system that predicts humidity levels and activates a mist maker when humidity drops below a threshold.	Temperature and humidity sensors, linear regression prediction model	Focuses only on humidity prediction. Does not estimate actual plant water requirements. Fertilization is not considered.

2. OBJECTIVE

2.1.Main Objective

The main objective of this research is to design and develop a smart IoT-based system that predicts watering needs and automatically manages watering and fertilization for Vanda orchids using multi-modal environmental data and ML techniques.

2.2.Specific Objectives

- To design and implement an IoT monitoring system that collects environmental data such as temperature, humidity, light intensity, and moisture levels around Vanda orchids.
- To collect and store environmental and historical plant data that can be used to analyze plant conditions and support ML model training.
- To analyze the relationship between environmental factors and orchid water requirements using the collected sensor data.
- To develop a ML model that predicts whether watering is required based on environmental conditions and historical data.
- To implement an automated irrigation system that activates a water pump when watering is predicted to be required.
- To implement an automated fertilization mechanism that determines fertilizer application using plant growth stage information and fertilization history, and supplies liquid fertilizer through a fertilizer pump when required.

3. METHODOLOGY

The proposed research follows a Design Science research approach combined with experimental validation. The design science approach is used to develop a technological solution for intelligent orchid care by integrating IoT monitoring, ML prediction, and automated irrigation control. Similar research has demonstrated that IoT-based irrigation systems can effectively monitor environmental conditions and automate water delivery using soil moisture and environmental sensors [1], [2]. In this study, environmental sensors will be used to collect data such as temperature, humidity, light intensity, and moisture levels around Vanda orchids. These sensors will transmit data to a microcontroller unit, where the collected information will be sent to a cloud platform for storage and processing.

The system will use ML classification techniques to predict whether watering is required based on environmental conditions and historical irrigation data. Research shows that IoT-based smart irrigation systems can improve water efficiency and crop productivity by making irrigation decisions based on real-time environmental data rather than fixed schedules [3]. In the proposed system, environmental features such as temperature, humidity, soil moisture, plant growth stage, time since last watering, and fertilization history will be used as input features for the prediction model. Several ML algorithms such as Decision Tree, Random Forest, and Logistic Regression will be evaluated, and the best performing model will be selected based on prediction accuracy.

For system implementation, ESP32 or ESP8266 microcontrollers will be used to collect sensor data and communicate with the backend system through Wi-Fi. Previous research has shown that microcontroller-based IoT irrigation systems using wireless communication can efficiently automate irrigation while reducing water waste and human intervention [1], [3]. The proposed system will also include automated actuation mechanisms, where a water pump will be activated when watering is predicted to be required. Fertilization decisions will be determined based on plant growth stage, fertilization history, and environmental conditions to ensure appropriate nutrient supply.

Environmental and historical plant data will be stored in a cloud database and used to train the ML model. The dataset will be divided into training and testing sets to evaluate model performance. Standard evaluation metrics such as accuracy, precision, recall, and F1-score will be used to assess the prediction performance of the model. Similar evaluation strategies have been used in IoT-based smart agriculture systems to validate irrigation decision models and ensure reliable automated irrigation control [2], [3].

4. HIGH-LEVEL SYSTEM ARCHITECTURE

1. Overview of the Proposed Solution and Its Components

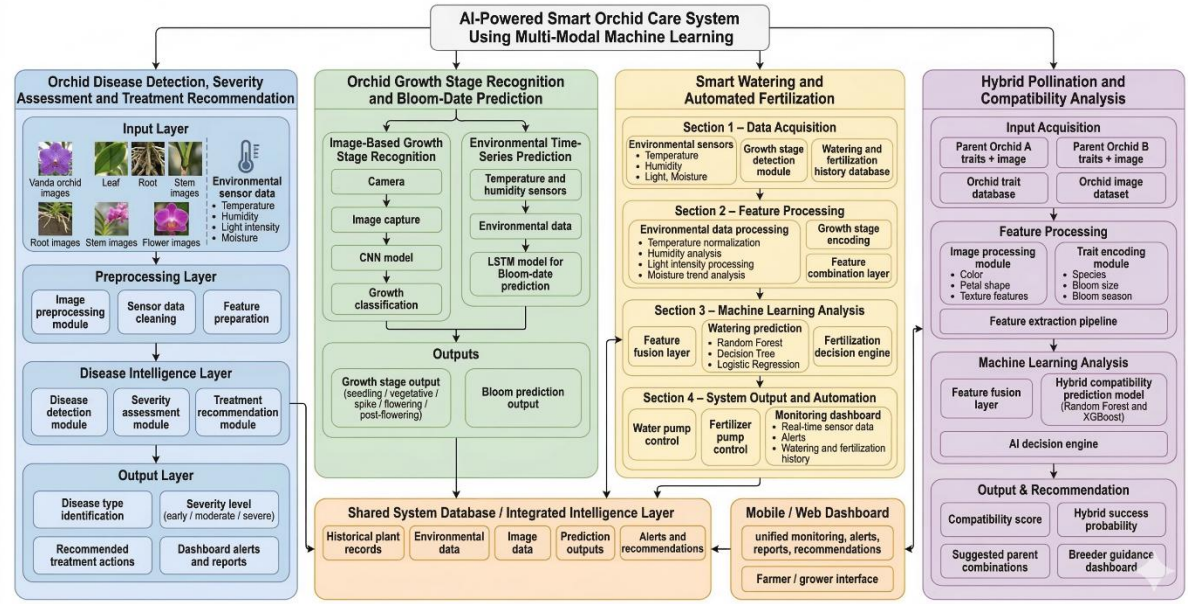


Figure 1 : High-Level System Diagram of AI-Powered Smart Orchid Care System Using Multi-Modal Machine Learning

The proposed AI-powered smart orchid care system is designed to support different aspects of orchid cultivation using intelligent technologies. The system integrates several components that work together to improve plant health, automate care processes, and support data-driven decision making for orchid growers.

The overall system consists of four main components: orchid disease detection and treatment recommendation, orchid growth stage recognition and bloom prediction, smart watering and automated fertilization, and hybrid pollination and compatibility analysis.

This research proposal focuses on the Smart Watering and Automated Fertilization component. Its main objective is to monitor environmental conditions around Vanda orchids, predict whether watering is required, and automatically control irrigation and fertilizer delivery. Proper water and nutrient management is essential for orchid health, root development, and flowering performance.

2. Focused Component: Smart Watering and Automated Fertilization

   The Smart Watering and Automated Fertilization component consists of several interconnected units. The environmental sensing unit collects real-time data such as temperature, humidity, light intensity, and moisture levels around the orchid environment.

   An ESP32 microcontroller acts as the IoT control unit, reading sensor data and sending it to the backend system through Wi-Fi. The backend system analyzes both real-time and historical data to determine whether watering is required.

   Based on this decision, the actuation unit activates the water pump or fertilizer pump automatically. A monitoring dashboard allows users to view sensor readings, predictions, and system activity in real time.

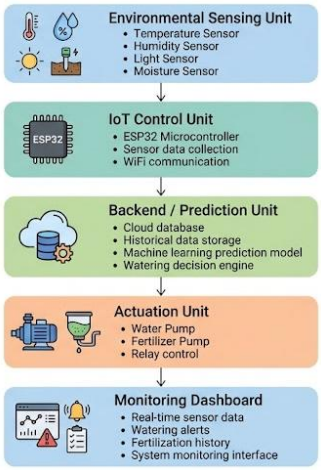


Figure 2: Component Architecture of the Smart Watering and Automated Fertilization Module

1. High-Level System Diagram

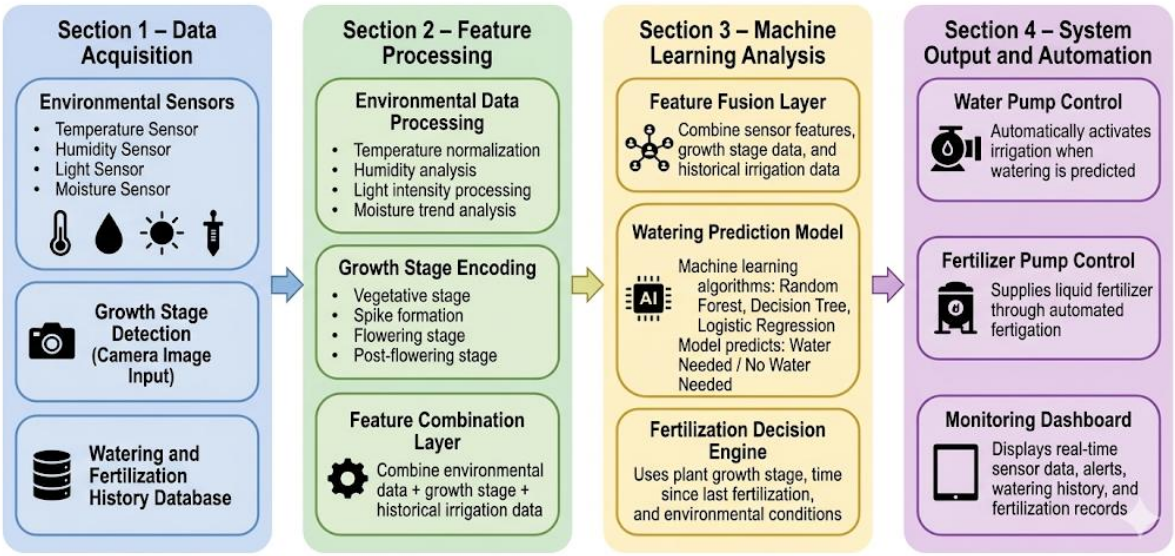


Figure 3 : High Level System Diagram Smart Watering and Automated Fertilization System for Vanda Orchids

2. Component Interactions

   The system operates through continuous interaction between sensors, the IoT device, the backend system, and automated pumps. Environmental sensors collect data from the orchid environment and send the readings to the ESP32 microcontroller. The ESP32 transmits the data to the backend system, where it is stored and analyzed.

   A ML model then predicts whether watering is required. If watering is needed, the system activates the water pump. Fertilization is determined based on plant growth stage, environmental conditions, and fertilization history, after which the fertilizer pump may be activated. All system activities and sensor readings are displayed on the monitoring dashboard for user observation.

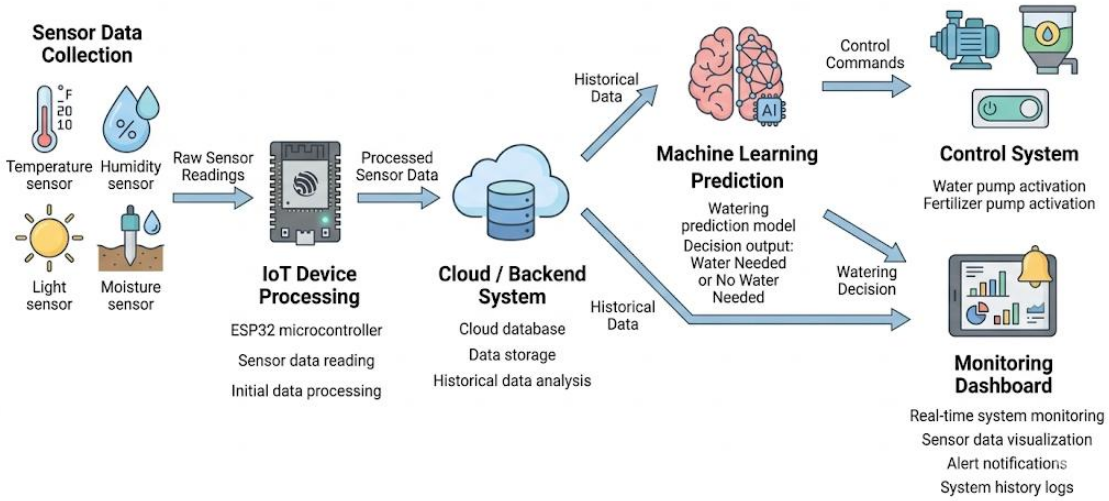


Figure 4 : Data Flow of the Smart Irrigation System

3. Integration with the Overall Project

   Although this research focuses on the Smart Watering and Automated Fertilization component, it supports the overall smart orchid care system. Maintaining proper water and nutrient levels helps create a stable growing environment, which improves plant health and supports other components such as disease detection and growth stage analysis. The system also generates historical plant care data that can support future research and decision making.

4. Scalability and Security Considerations

   The proposed architecture is designed to be scalable and secure. Additional sensors can be added to monitor more environmental parameters, and the system can be expanded to support multiple orchids or larger cultivation environments.

   To ensure security, sensor data should be transmitted using secure communication methods, user access to the monitoring dashboard should be authenticated, and stored data should be protected from unauthorized access.

5. USER REQUIREMENTS

1. Stakeholder Identification
   Several stakeholders interact with the proposed smart orchid care system.
   - Orchid Growers
     These are the primary users of the system. They use the system to monitor environmental conditions and automate watering and fertilization for Vanda orchids.
   - Home Gardeners
     Individuals who grow orchids at home can use the system to maintain healthy plants without needing expert knowledge about irrigation and fertilization.
   - Agricultural Experts
     Agricultural Experts may use the collected environmental data to study orchid growth patterns and improve smart agriculture technologies.
   - System Administrators
     Administrators maintain the system, manage the database, and ensure the system operates correctly.
2. Functional Requirements
   The functional requirements describe the main operations that the system must perform.
   - FR1**:** The system shall collect environmental data such as temperature, humidity, light intensity, and moisture levels using sensors.
   - FR2**:** The system shall store environmental data in a database for monitoring and analysis.
   - FR3**:** The system shall analyze environmental data to determine whether watering is required.
   - FR4**:** The system shall automatically activate the water pump when watering is needed.
   - FR5**:** The system shall automatically activate the fertilizer pump when fertilization is required.
   - FR6**:** The system shall provide real-time monitoring of environmental conditions through a dashboard.
   - FR7**:** The system shall record watering and fertilization events in the system database.
   - FR8**:** The system shall notify users when watering or fertilization actions occur.
3. NFRs
   Non-functional requirements describe system performance, reliability, and usability expectations.
   - Performance Requirements

- NFR1**:** The system should update sensor readings at regular intervals (for example every 5 minutes).
- NFR2**:** The system should process environmental data and determine watering decisions within a few seconds.

- Reliability Requirements
  - NFR3**:** The system should operate continuously without interruption during plant monitoring.
  - NFR4**:** The system should maintain reliable communication between sensors, IoT device, and the backend system.
- Usability Requirements
  - NFR5**:** The monitoring dashboard should present environmental data in a clear and easy-to-understand format.
  - NFR6**:** Users should be able to view watering history and system status easily.
- Security Requirements
  - NFR7**:** Access to the monitoring dashboard should be protected using user authentication.
  - NFR8**:** Environmental data stored in the system database should be protected against unauthorized access.
- Scalability Requirements
  - NFR9**:** The system should support expansion to monitor multiple orchids or multiple sensor units in the future.

6. COMMERCIALIZATION PLAN

1. Target Market and Customer Persona
   The proposed smart orchid care system is designed mainly for orchid growers and plant enthusiasts who want to improve plant health and reduce
   The main target users are:

   - Orchid Nurseries
     Commercial orchid farms that grow orchids for sale can use the system to automate watering and fertilization. This helps maintain consistent plant health and reduce labor costs.
   - Home Orchid Growers
     People who grow orchids at home often struggle to maintain proper watering schedules. The system helps them maintain healthy orchids without needing expert knowledge.
   - Agricultural Researchers
     Researchers studying smart agriculture and plant monitoring can use the collected environmental data for research and analysis.

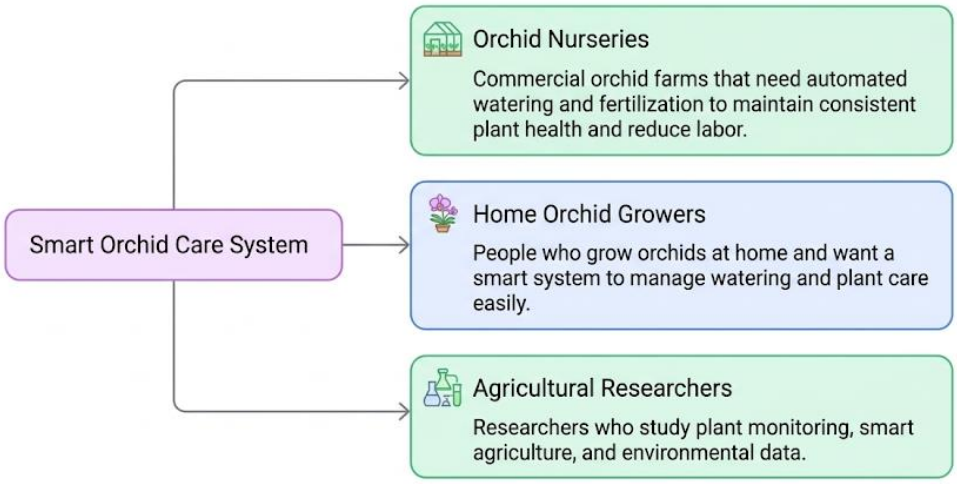


Figure 5 : Target Market Segments for the Smart Orchid Care System

2. Value Proposition
   - The proposed system provides several benefits to orchid growers:
     - Automated watering and fertilization, reducing manual work
     - Improved orchid health through data-driven plant care
     - Efficient water usage, reducing water waste
     - Better plant monitoring using real-time environmental data
     - Support for smart agriculture practices
     - Compared with traditional orchid care methods, the system provides intelligent plant management using sensors and ML.

3. Revenue Model

- The system can generate revenue through several approaches.
  - Hardware Device Sales
    Users can purchase the smart orchid monitoring device that includes sensors, a microcontroller, and irrigation control modules.
  - Software / Monitoring Platform
    A mobile monitoring platform can be offered with advanced features such as plant analytics and historical data reports.
  - Subscription Services
    Advanced plant analytics, predictive insights, and cloud storage could be offered as a subscription-based service.

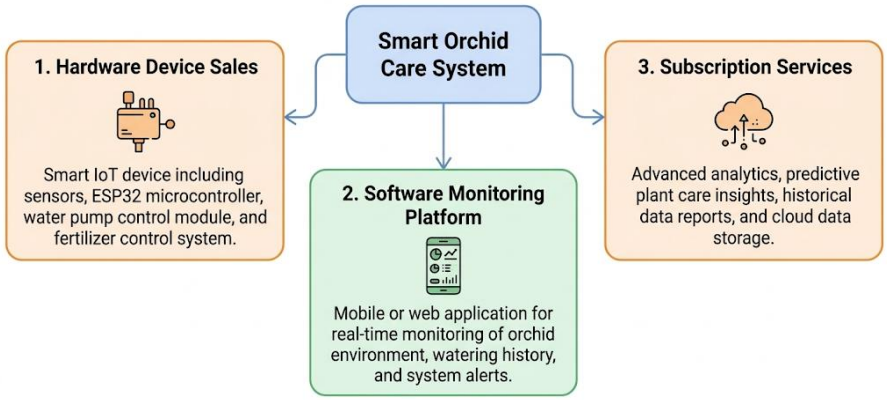


Figure 6 : Revenue Model of the Smart Orchid Care System

4. Cost Estimation

Table 2 : Cost Estimation for the Smart Watering and Automated Fertilization System

Item	Amount (LKR)	Description
Raspberry Pi Camera Module	4500 - 6000	Used to capture plant images
DHT22 Temperature and other Sensors	900 - 1200	Measures environmental conditions
ESP32 Microcontroller	2000 - 3000	Collects sensor data and controls the system
Raspberry Pi Zero 2W	3500 - 4500	Used for image processing
Power Supply and Cables	2500 - 3500	Required hardware connections
Total Hardware Cost per Unit	13400 - 18200	Estimated cost for one system
Software Tools	0	Python and ML libraries (open source)
Cloud Services	0	Firebase free tier for data storage
Estimated Total Cost	**13400 - 18200**	**Approximate system cost**

Cost Per Farmer (to set up one customer)
Hardware - LKR 18,200
Installation and travel – LKR 2,000
Total Per Farmer - LKR 22,200

5. Pricing Strategy

   The proposed smart orchid care system can generate revenue through both hardware device sales and subscription-based monitoring services.

   Users will first purchase the hardware device, which includes sensors, a microcontroller, and the irrigation control system. After installation, users can access the monitoring dashboard and smart watering features through a subscription plan.

   Two simple pricing plans are proposed.

Table 3 : Pricing Strategy

Plan	Monthly Price (LKR)	Features
Basic Plan	500	Environmental monitoring and watering alerts
Premium Plan	1,000	Smart watering, automated fertilization, and monitoring dashboard

6. Competitive Advantage

   - The proposed system offers several advantages compared with existing solutions.
     - Uses ML to support smarter watering decisions
     - Integrates both watering and fertilization automation
     - Designed specifically for orchid cultivation
     - Provides real-time monitoring and historical data analysis

     Most existing plant irrigation systems only automate watering based on simple sensor thresholds. The proposed system introduces predictive plant care and integrated nutrient management, which improves overall plant maintenance.

7. BUDGET AND JUSTIFICATION

The following table shows the estimated budget required to develop the AI Powered Smart Orchid Care System Using Multi-Modal ML.

Table 4 : Budget and Justification

Hardware Costs (per unit)			
Cost Category	Estimated Amount (LKR)	Notes	Justification
Raspberry Pi Camera Module	4,500–6,000	Captures plant images	Captures daily images of Vanda orchids to track growth stages (seedling, vegetative, spike, flower, post-bloom).
Temperature sensor, Humidity sensor, Light sensor, Moisture sensor	900–1,200	Measures environment	Measures greenhouse environment.
ESP32 Microcontroller	2,000–3,000	Processes sensor data	Reads sensor data and sends it to the cloud via WiFi. Low cost and low power consumption.
Raspberry Pi Zero 2W	3,500–4,500	Runs image processing	Runs the image processing at the edge. It captures and analyzes images before sending results to the cloud.
Power supply, cables, SD card	2,500–3,500	Basic components	Provides stable power to the Raspberry Pi and sensors. Stores the operating system and temporary image data. Connects sensors to the board.
3D printed enclosure	1,000–1,500	Protects electronics	Protects electronics from greenhouse humidity, dust, and heat.
Total Hardware per Unit	14,400–19,700		
Software Costs			
Cost Category	Estimated Amount (LKR)	Notes	Justification
All software (Python, TensorFlow, etc.)	0	Open source	Free, open-source
Total Software	0		
Cloud Services (monthly)			
Cost Category	Estimated Amount (LKR)	Notes	Justification
Firebase free tier	0	Up to 1 GB storage	Provides database and storage for free up to 1GB, which is enough for a pilot project with one unit.
Total Cloud (Year 1)	0		
Human Effort Costs			
Cost Category	Estimated Amount (LKR)	Notes	Justification

Travel to greenhouses (fuel, bus fare)	5,000–8,000	10–15 visits	
Mobile data/Wi-Fi charges	5,000–8,000	For cloud testing and updates	
Refreshments during fieldwork	4,000–6,000	Snacks/meals during long days	
Total Human Effort	14,000–22,000		

8. WORK BREAKDOWN STRUCTURE (WBS)

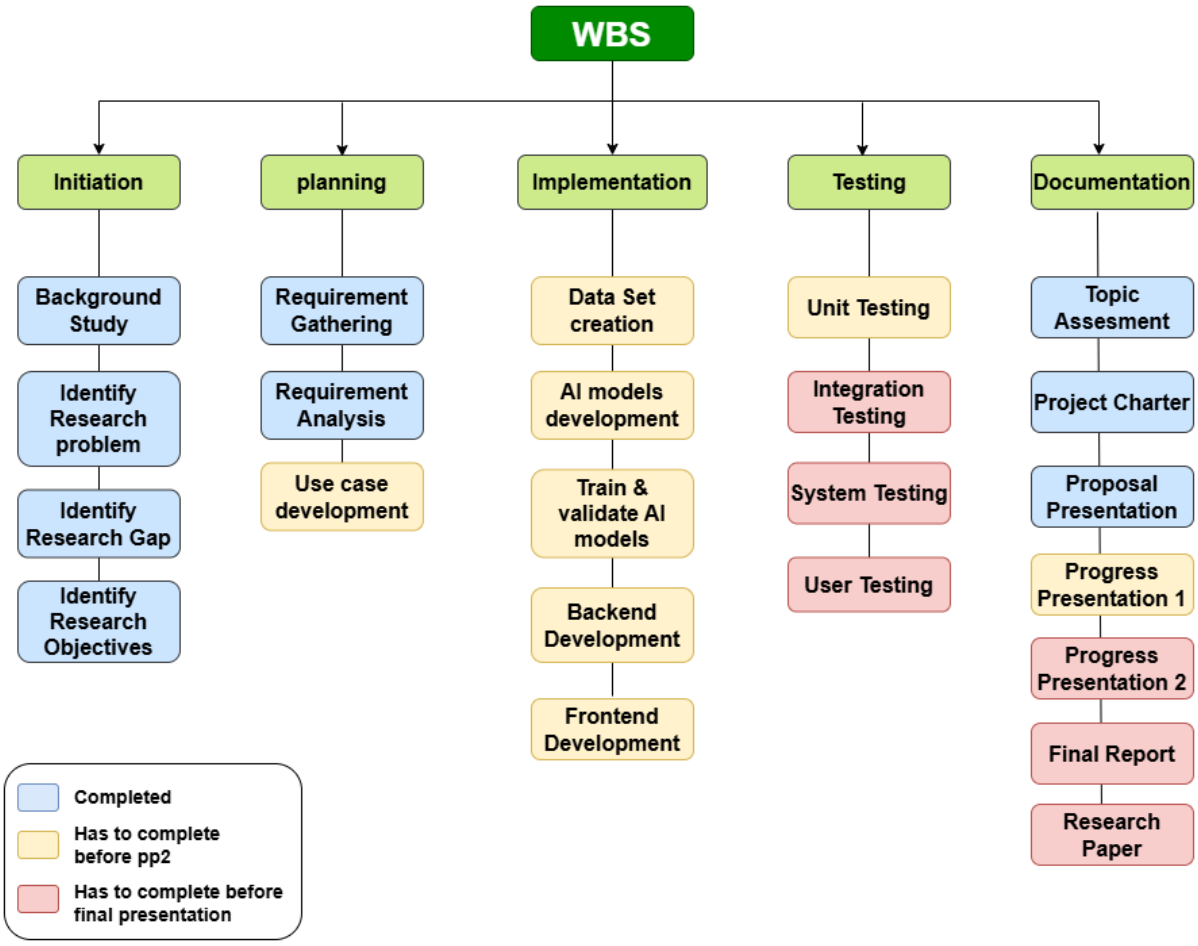


Figure 7 : Work Break Down Structure

9. GANTT CHART

Table 5 : Gantt chart

	Semester 1						Semester 2					
Activity	1	2	3	4	5	6	7	8	9	10	11	12
Preparation												
Identify the Problem												
Requirement gathering												
Literature review												
Analyze requirements												
Online project registration												
Topic assessment from submission												
Identify main objectives												
Project proposal submission												
Data Collection												
Model Training												
Software Development												
Progress presentation 1												
Integration												
Testing												
Progress presentation 2												
Research paper submission												
Progress presentation 2												
Finalizing												

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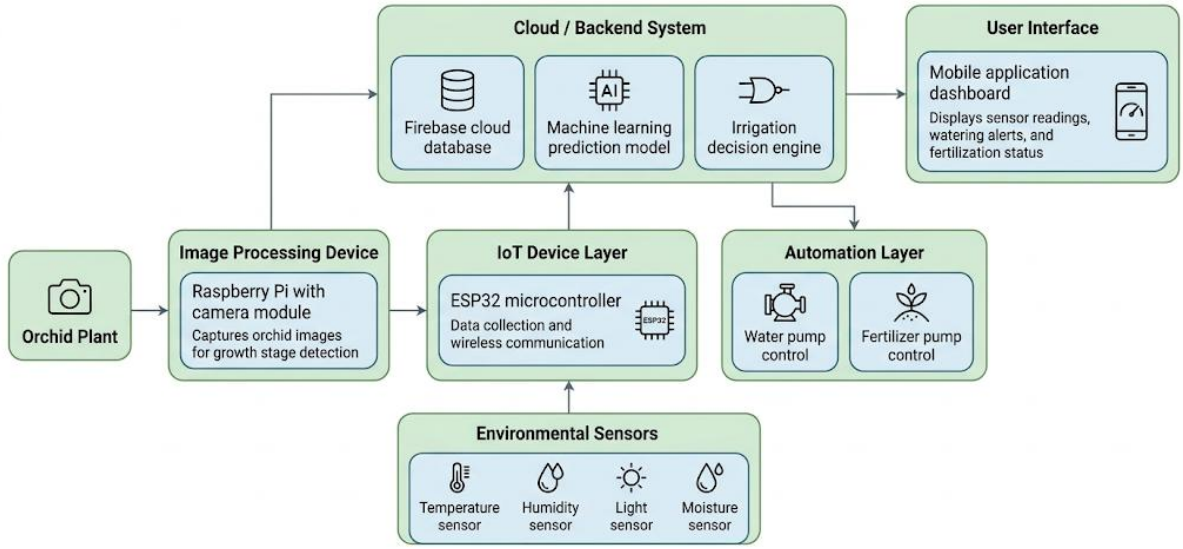
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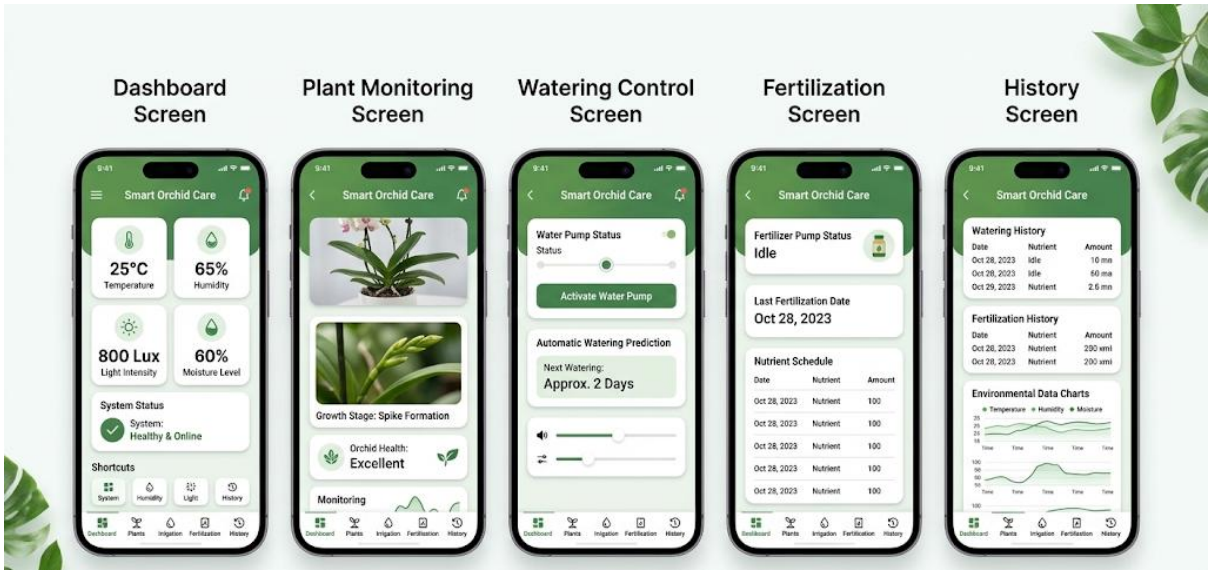
11. APPENDICES

- Detailed System Architecture



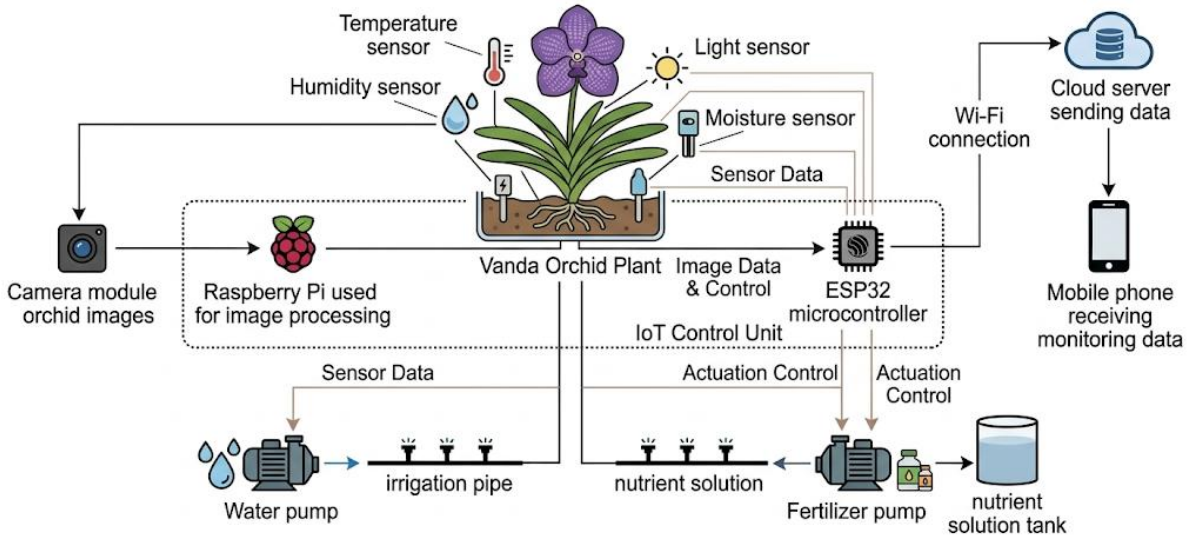
Appendix A : Detailed System Architecture

- Mobile App UI Mockups



Appendix B : Mobile App UI Mockups

- IoT Hardware Setup



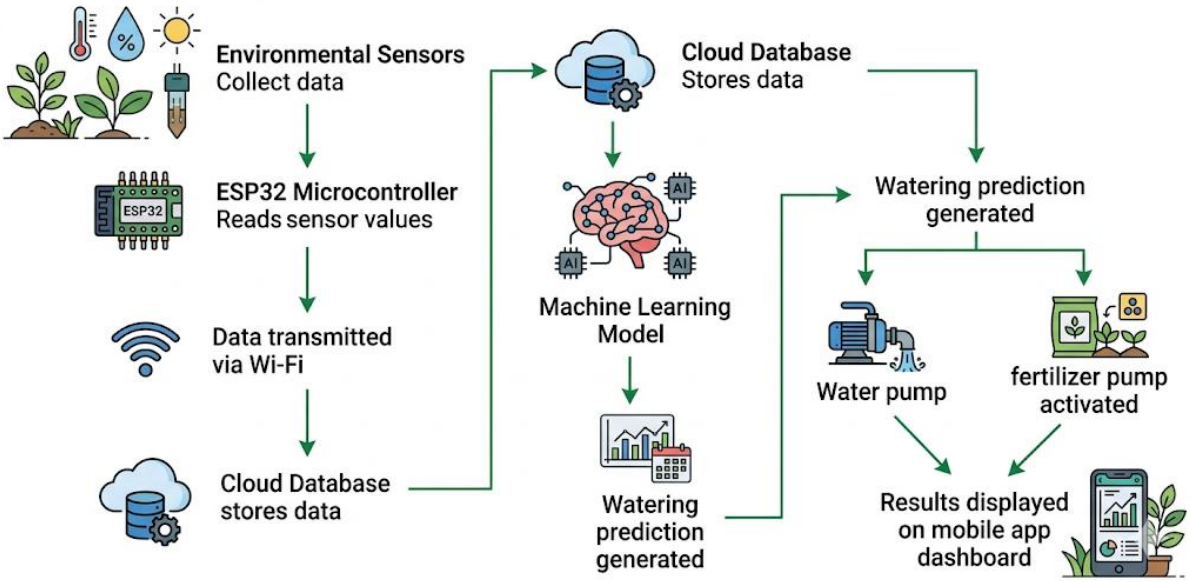
Appendix C : IoT Hardware Setup

- Risk Analysis Diagram

Risk	Mitigation Strategy
Sensor failure	Sensor calibration
Network connectivity issues	Offline data storage
Overwatering risk	Safety watering limits
Power supply failure	Backup power supply
Incorrect prediction by machine learning model	Model validation and testing

Appendix D : Risk Analysis Diagram

- Data Collection Process



Appendix E : Data Collection Process